A-A-51867A
12 July 1991
SUPERSEDING
A-A-0051867(DM)
20 April 1987

COMMERCIAL ITEM DESCRIPTION

ELECTROSURGICAL APPARATUS

The General Services Administration has authorized the use of this commercial item description

This commercial item description covers an electrosurgical apparatus consisting of a solid state, radio frequency generator, operator controls and applicator devices

Salient characteristics:

1 The apparatus shall be a solid state electrosurgical unit (ESU) with monopolar and bipolar capability, suitable for use in surgical procedures

2 Monopolar settings shall include a pure cut, pure coagulation, and a blend of cut and coagulation Power output of the ESU shall remain relatively constant over the range of load impedance of 100 to 1000 ohms In no instance shall maximum power exceed 400 watts unless clearly stated on a label attached to the ESU and in the operator documentation

a In a pure cut mode, the monopolar output power shall provide a minimum range of 0 to 300 watts, as measured into a manufacturer-specified resistive load that must be between 100-500 ohms, inclusive The unit shall be capable of delivering at least 150 watts of cutting power into 500 ohms

b In the coagulation mode, the monopolar output power shall provide a minimum range of 0 to 100 watts, as measured into a manufacturer-specified resistive load that must be between 100-500 ohms, inclusive The unit shall be capable of delivering at least 50 watts of coagulation power into 500 ohms and exhibit minimal cutting characteristics when operating in this mode

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:02Z
Check the source to verify that this is the current version before use.

c The blend mode shall provide cutting waveforms with a choice of at least three degrees of hemostasis, or allow variation of the relative amounts of hemostasis and cutting at any power setting by means of stepped or stepless controls

3 Bipolar output control shall be independent of monopolar controls, and shall produce a coagulation waveform Bipolar output power shall be adjustable and provide a minimum range of 0 to 30 watts, as measured into a manufacturer-specified resistive load that must be between 50-300 ohms, inclusive. Maximum output power shall not exceed 100 watts, unless a label clearly states otherwise

4 Open circuit voltage of the unit shall be high enough to allow the surgeon to fulgurate (spray coagulate) tissue In the coag mode, open circuit voltage, peak-to-peak, shall be at least 2200 volts into a 500 ohm load The unit shall withstand repeated and prolonged open circuit voltage

5 The electrosurgical apparatus shall have separate, adjustable cut, coagulate, and bipolar output power level controls The controls may be stepped or continuously variable At the minimum power setting, there shall be zero output power at zero control setting If the unit has no zero setting, the minimum output power at any load resistance between 100-1000 ohms shall not exceed five percent of the rated maximum output power for that mode Power increase shall not exceed 80 watts or 50 percent, whichever is greater, for each step of a stepped control or 10 percent advance in rotation of a continuously adjustable control

The use of momentary activation switches for selection of power level is acceptable If momentary activation switch is provided, the unit must be set to zero power level or "standby" mode at initial power up

6 Output shall be activated by means of a footswitch, a hand control style electrode pencil, and forceps with a built-in switch

a Footswitch activation Footswitch shall be splashproof, grounded, and have suitable mechanical protection against accidental depression. Shall be designed so that pure cut mode output is activated by the leftmost control of the footswitch as viewed by the operator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:02Z
Check the source to verify that this is the current version before use.

b Handpiece actuation The hand control style electrode pencil shall provide the capability for the user to select between the coagulation and cutting modes at the handpiece in the monopolar mode If two switches are provided, the switch nearer the active electrode shall activate the cutting mode The 'cut' switch shall be color coded yellow and the 'coagulation' switch shall be color coded blue With rocker or sliding types of switch mechanisms, a forward motion (toward the electrode) or pressure on the forward part of the switch mechanism shall activate the cutting mode

c Forceps actuation Monopolar or bipolar outputs shall be switch-selectable by means of a switch built into the forceps

7 If the cutting and coagulation switches are actuated simultaneously, the unit shall operate in the coagulation mode or not at all, unless it is designed to produce a blended output under this condition

8 Output power, as measured at the active electrode, shall remain constant within 10 percent over the full tolerance range of line voltages specified in paragraph 10

9 The unit shall have isolated output circuits.

10 The unit shall generate minimal radio frequency interference (RFI), both radiated and conducted, through the power line The active cable of the ESU shall be properly insulated to limit radiated losses and prevent capacitive coupling into nearby wires and metal equipment

11 The unit shall be provided with a positive or self-latching on/off switch An indicator light shall signal that the ESU has been turned on Separate visual indicators shall be incorporated to indicate when the ESU 'RF' output has been activated In addition, a minimum of two distinct audible tones shall indicate cut, and coag/bipolar output activation An adjustable volume control which cannot reduce the sound level below 45 dB shall be provided

12. Front panel controls shall be visible and easily identifiable from a distance of at least 2 5 meters (8 feet) for vision corrected to 20/20 Footswitch and electrode connections shall be clearly labeled, color coded, or size keyed to prevent accidental attachment to improper outlets Control indicators shall be color coded Controls and output connectors shall be arranged in an orderly sequence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:02Z
Check the source to verify that this is the current version before use.

Condensed operating instructions and warning placards shall be mounted on the unit

13 To prevent patient burns, the dispersive electrodes (disposable and reusable) shall disperse the return current over an adequately large area A reusable dispersive electrode shall be supplied which shall resist scratching and be easy to clean The electrodes shall not be adversely affected by surgical prep solutions or physiological fluids

14 The unit shall have a dispersive electrode monitoring system to monitor the contact quality between the patient and the dispersive electrode

15 A detection system shall be provided to monitor the electrical continuity of the current path from dispersive electrode to the ESU If a break occurs in this pathway, the ESU shall de-energize and activate a fault indicator until the unit is manually reset for normal operation

16 The unit shall be convection cooled A fan, which must be thermally activated, may be incorporated to provide adequate component cooling and to dissipate internal heat If a fan is provided placement of ventilation openings shall deflect exhaust air downward and away from the sterile field Position of vents shall not allow accidental contact with electrical connections or components, and shall be designed to restrict entry of spilled fluids

17 Overcurrent protection for the ESU, preferably in the form of a circuit breaker, is required. If a fuse is used, a spare must be provided and clearly labeled as to function and rating Circuit breakers must be protected from other objects impinging against it

18 Unit shall meet UL 544 and ANSI/AAMI HF-18 Standards All items shall conform to the applicable requirements of nationally recognized standards for the safety of medical equipment of the issue in effect on the date of solicitation All items shall be approved as conforming to such a standard by an independent, nationally recognized testing laboratory

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:02Z
Check the source to verify that this is the current version before use.

19 Electrical Requirements·

a The electrosurgical apparatus shall be capable of operating over the full tolerance range of one or more of the following line voltages and frequency combinations as specified by the customer requisition·

1 115 VAC plus or minus 10 percent, 50 plus or minus 2 Hz
2 115 VAC plus or minus 10 percent, 60 plus or minus 2 Hz
3 230 VAC plus or minus 10 percent, 50 plus or minus 2 Hz
4 230 VAC plus or minus 10 percent, 60 plus or minus 2 Hz.

The dual voltage and hertz feature is desirable but not mandatory

b Change over devices, switches, circuits, or transformers if used for the purpose of voltage or frequency conversion, shall be clearly marked and identified in the appropriate documentation which accompanies the unit Mechanical or electrical adjustments to effect the conversion shall not require the use of specialized tools or skills

20 Minimum length of the cords and cables supplied shall be·

Cable Type	Length
Line Cord	3 0 meters (9 8 feet)
Active	2 5 meters (8 2 feet)
Dispersive	2 4 meters (7 9 feet)
Footswitch	3 0 meters (9 8 feet)

2. Each unit shall be supplied with all accessories necessary for a fully functional system As a minimum, the following accessories shall be included·

a One mobile cart Cart shall provice storage space for the footswitch and other accessories When the ESU is mourtec on the cart, the front panel controls shall be highly visible anc easily identifiable

b Twelve disposable dispersive electrodes with cord

c Twelve disposable dispersive electrodes, split plate, full surface conductor, with preattached cord.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:02Z
Check the source to verify that this is the current version before use.

d One dispersive electrode adapter, if necessary

e One reusable dispersive electrode

f One disposable hand activated handle, presterilized, having a one inch blade, cord and connector

g One reusable hand activated handle, presterilized, which accepts standard 3/32 inch shaft electrodes, having a removable one inch blade, cord, and connector.

h One reusable foot activated handle, which accepts standard 3/32 inch shaft electrodes, with cord

i One bipolar forceps, hand switching, 6-1/2 inch or 7 inch, straight tip, with cord

j One forceps, Hudson (Ewald), switching bipolar, 4-3/4 inch straight tip

k One (or two) footswitch(es), for monopolar and bipolar activation

l Twelve tubes electrosurgical gel, 5 ounces per tube

m Twelve electrodes, 1 inch, straight blade

n One electrode 7/32 inch, ball

o One electrode, 3/4 inch or 1 inch, needle

p One electrode 3/32 inch, angled ball

q One electrode, 3/8 inch, round loop

22 Operation and Service Data Manuals for Medical and Laboratory Instrumentation

a Information Content

The contractor shall furnish manuals, handbooks and/or brochures containing complete operation, installation and service/maintenance instructions (including pictures or illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures will be written in the English language and all schematics and wiring diagrams will use American electrical and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:02Z
Check the source to verify that this is the current version before use.

electronics symbols The manuals will include electrical data and connection diagrams for all applicable utilities The instructions shall also contain a complete list of all replaceable parts showing part number, name, and quantity required All service notes, service memos, etc pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included These instructions shall be, as a minimum, that furnished to service engineers (or servicemen) that normally install and service the equipment for the company or distributor. When the system being procured includes, or will operate ancillary equipment the service data shall include complete instructions and drawings which show interfacing or all system components The operator's/instruction manual shall include a listing of cautionary warnings of the potential hazards of electrosurgical devices, including electric shock, thermal burns, fire and explosion, cataract formation, pacemaker interference, RF - interference and irradiation, cardiac arrhythmias, and allergic or toxic reaction to materials The operator's manual shall also state the recommended method of sterilization of all required sterile reusable components

b Quantities, Distribution and Schedules·

Two (2) complete sets of the above information shall be packaged with each unit of the required equipment Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, then only a total of 2 service manuals need to be supplied to each consignee However, 2 operation manuals still need to be supplied with each unit

Workmanship. The electrosurgical apparatus shall be free from defects which detract from its appearance or impair its serviceability

Unit. Each (ea) One electrosurgical apparatus as specified, constitutes one unit

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:02Z
Check the source to verify that this is the current version before use.

Quality Assurance Provisions

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

First article inspection. First article inspection shall be performed on the electrosurgical unit when a first article sample is required. This inspection shall include examination and tests.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of S-2. The unit of product for examination purposes shall be one electrosurgical apparatus.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of S-2. The unit of product for test purposes shall be one electrosurgical apparatus.

Tests. Tests shall be conducted to determine compliance with specification requirements. The entire unit shall be assembled with all components properly connected. Testing shall include but shall not be limited to the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:02Z
Check the source to verify that this is the current version before use.

Input power test. Verify the unit operates on each specified voltage and Hz All remaining tests are to be performed using each specified input power

Output power test. Verify the following:

a The monopolar output power,at any setting, peaks at a load impedance between 100 and 1000 ohms

b In a pure cut mode, the monopolar output power provides a minimum range of 0 to 300 watts as measured into a resistance load specified by the manufacturer (must be between 100 - 500 ohms)

c The unit is capable of delivering at least 150 watts of cutting power into 500 ohms

d In the coagulation mode, the monopolar output power provides a minimum range of 0 to 100 watts as measured into a resistance load specified by the manufacturer (must be between 100 - 500 ohms)

e The unit provides coagulation power of a least 50 watts and exhibits minimal cutting characteristics when operating in this mode

f In the bipolar mode, output power provides a minimum range of 0 to 30 watts as measured into a resistance load specified by the manufacturer (must be between 50 - 300 ohms)

Maximum output power test. Verify the maximum power in monopolar settings into any load resistance does not exceed 400 watts, unless a label clearly states otherwise Verify that maximum output in bipolar settings does not exceed 100 watts unless a label clearly states otherwise

Power control sensitivity test. Verify the power increase does not exceed 60 watts or 50 percent, whichever is greater, for each step of a stepped control or 10 percent advance in rotation of a continuously adjustable control

Open circuit voltage test. Verify that in the coag mode, the open-circuit voltage, peak to peak, is at least 2200 volts into a 500 ohm load

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:02Z
Check the source to verify that this is the current version before use.

Minimum power control setting test.

a For units with stepped or continuously variable power level controls verify at the minimum power setting there is zero output power at the zero control setting. If the unit has no zero setting, verify the minimum output power at any load resistance in the range of 100 to 1000 ohms does not exceed 5 percent of the maximum output power for that mode

b For units with momentary activation switches, verify the unit is set to zero power level or 'standby' mode at initial power-up

Line voltage fluctuation test Verify that the output power measured at the active electrode remains constant within 10 percent when the line voltage is fluctuated over the tolerance range specified for each voltage/frequency combination specified under salient characteristics paragraph 1B

Visual Indicators test. Verify the unit provides separate visual indicators for input power turned on and for output activated

Audible indicators test Verify the unit provides distinct audible tones to indicate cut and coag/bipolar activation. Verify the volume can not be reduced below 45 dB

Output isolation test Verify the power delivered to a 200 ohm load connected between each active output and ground does not exceed 4 5 watts at maximum output power setting

Radio frequency leakage test. Verify the power delivered to a 200 ohm non-reactive resistance connected from the dispersive electrode to ground does not exceed 4 5 W at maximum output power setting with the electrodes operated open-circuited

Electrical safety test. Verify no more than 500 uA of sink current flows to ground with 110 V, 60 Hz applied to the active or return electrode

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:02Z
Check the source to verify that this is the current version before use.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable

The contracting officer has the option of accepting or rejecting the product

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract

Regulatory requirements.

Federal Food, Drug and Cosmetic Act If the product covered by this document has been determined by the U S Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder In addition, the offeror/contractor shall comply and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations

Recovered materials The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 as amended, and as implemented by Federal Acquisition Regulation Subpart 23 4 to the maximum extent practical

Preservation, packaging, packing, labeling and marking. Unless otherwise specified, preservation packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc , involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:02Z
Check the source to verify that this is the current version before use.

Ordering data Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order

NOTE The following National Stock Number is covered by this document:

6515-01-314-4283

MILITARY INTERESTS

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No 6515-4158

Location ESUMC17 DOC

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:02Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: A-A-51867A	2. DOCUMENT DATE (YYMMDD): 12 July 1991

3. DOCUMENT TITLE: ELECTROSURGICAL APPARATUS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial: AC 215-737-2870 (2) AUTOVON: AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19101-8419	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 — Previous editions are obsolete — 96/PPC

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:02Z
Check the source to verify that this is the current version before use.